



Module
Presenter's Manual
for
**Server-side
Development**

***Effective from: March 2024
Ver. 1.0***

Amendment Record

Version No.	Effective Date	Change	Replaced Pages
1.0	March 2024	New	-

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1. Introduction

At the end of this course, students will be able to:

- Describe Node.js, its features, and its installation process.
- Describe the architecture of Node.js.
- List benefits and limitations of Node.js applications.
- Describe built-in modules in Node.js.
- Describe the process of uploading files using Node.js.
- Explain the concept of local Modules and their application in Node.js.
- Identify the advantages and disadvantages of RESTful and HTTP APIs.
- Describe the steps to connect the Web application to a MongoDB database.
- List the steps to upload the Web application to the GitHub repository.

2. Information on Session Allocation

Module	Online (No. of Hours)	Self-study (No. of Hours)
NODEJS	24	12

Throughout this Presenter's Manual, the module **Server-side Development** will be referred to as **NODEJS**.

3. Week-wise Session Schedule

- A session is of 2 hours duration.

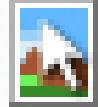
➤ Week-Wise Schedule

Week	Day 1	Day 2		Day 3	Day 4	
1	Session 1 NODEJS – TL1	Session 2 NODEJS – TL2		Session 3 NODEJS – TL3	Session 4 NODEJS – TL4	
2	Session 5 NODEJS – TL5	Session 6 NODEJS – TL6	S1-S3	Session 7 NODEJS – TL7	Session 8 NODEJS – TL8	
3	Session 9 NODEJS – TL9	Session 10 NODEJS – TL10		Session 11 NODEJS – TL11	Session 12 NODEJS – TL12	S4-S6

TL: Online Session

S: Self-study Hours (over and above the module duration)

Resources available on OnlineVarsity for Students:

Icons	Feature - Description/Functionality
	Download Book - Student has the option to download the subject related e-book and read offline.
	Glossary - Student can access a list of subject related specialized words with their definitions.
	FAQ - Student can access frequently asked questions and their answers.
	Practice 4 Me - Student can test and evaluate their understanding of module related topics.
	Work Assignments - Student can solve scenario based lab assignments (Hands-on). The faculty will evaluate and give their feedbacks.
	References - Student can access additional subject related material for reading.
	Feedback - Student can provide feedback on the course material.
	Ask to Learn – Student can submit subject related technical queries. Queries submitted will be directed to the particular course coordinator/head.

4. Session Coverage

Session No.	Session Title	Session Details	Deliverables' Mapping
1	NODEJS – TL1	All the topics as listed below from Session 1 of <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 1 – Introduction to Server-side Development</u> ➤ Outline the client-server architecture ➤ Explain the significance of server-side programming ➤ Explain a Web server and its architecture ➤ Describe Node.js, its features, and its installation process	<u>Server-side Development with NodeJS</u> SG – Session 1 XP – Session 1 TG – Session 1
2	NODEJS – TL2	All the topics as listed below from Session 2 of <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 2 – Node.js Essentials</u> ➤ Describe the architecture of Node.js ➤ List benefits and limitations of Node.js applications ➤ Explain the concept of console-based applications in Node.js ➤ Explain major programming constructs in Node.js ➤ Compare the roles of Node Package Manager (NPM) and Node Version Manager (NVM) in Node.js ➤ Describe global objects and their use in Node.js applications	<u>Server-side Development with NodeJS</u> SG – Session 2 XP – Session 2 TG – Session 2
3	NODEJS – TL3	All the topics as listed below from Session 3 of <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 3 – Modules and Packages in Node.js</u> ➤ Explain what modules are and its three types ➤ Explain the significance of	<u>Server-side Development with NodeJS</u> SG – Session 3 XP – Session 3 TG – Session 3

Session No.	Session Title	Session Details	Deliverables' Mapping
		packages ➤ Describe the purpose of Web framework and utility functions	
4	NODEJS – TL4	All the topics as listed below from Session 4 of <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 4 – Built-in Modules</u> ➤ Describe built-in modules in Node.js ➤ Explain http module to create an HTTP server and how to use it ➤ Explain how to use the url module to parse and work with URLs ➤ Explain how to use the events module to handle asynchronous events	<u>Server-side Development with NodeJS</u> SG – Session 4 XP – Session 4 TG – Session 4
5	NODEJS – TL5	All the topics as listed below from Session 5 of <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 5 – More Built-in and Local Modules</u> ➤ Explain the purpose and usage of querystring in Node.js ➤ Describe how to send emails using Node.js ➤ Describe the process of uploading files using Node.js ➤ Explain the concept of local Modules and their application in Node.js	<u>Server-side Development with NodeJS</u> SG – Session 5 XP – Session 5 TG – Session 5
6	NODEJS – TL6	The Try It Yourself questions from Session 1 to Session 5 of <i>Server-side Development with NodeJS</i> book should be covered in this session.	<u>Server-side Development with NodeJS</u> Session 1-5
	S1-S3	All the Assignments from <i>Session 1 to Session 5 of Server-side Development with NodeJS</i> course should be solved in this session.	<u>Server-side Development with NodeJS (OnlineVarsity)</u> Session 1-5

Session No.	Session Title	Session Details	Deliverables' Mapping
7	NODEJS – TL7	All the topics as listed below from Session 6 of <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 6 – Introduction to Express.js Framework</u> ➤ Describe the Express.js framework ➤ Explain the architecture of Express.js ➤ List the advantages of using Express.js ➤ Explain the working of Express.js	<u>Server-side Development with NodeJS</u> SG – Session 6 XP – Session 6 TG – Session 6
8	NODEJS – TL8	All the topics as listed below from Session 7 of <i>Server-side Development with NodeJS</i> book should be covered in <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 7 – Asynchronous and Synchronous Programming</u> ➤ Compare asynchronous and synchronous programming models ➤ Explain asynchronous programming model using callback, Promise, and async/await in Node.js ➤ Describe synchronous programming model in Node.js ➤ Explain the event loop architecture in Node.js	<u>Server-side Development with NodeJS</u> SG – Session 7 XP – Session 7 TG – Session 7
9	NODEJS – TL9	All the topics as listed below from Session 8 of <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 8 – RESTful and HTTP APIs</u> ➤ Describe an API and its purpose ➤ List the common types of APIs in Node.js and their functions ➤ Explain RESTful and HTTP APIs and their features	<u>Server-side Development with NodeJS</u> SG – Session 8 XP – Session 8 TG – Session 8

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain how to use RESTful API and HTTP API to manage data ➤ Identify the advantages and disadvantages of RESTful and HTTP APIs ➤ Distinguish between RESTful APIs and HTTP APIs	
10	NODEJS – TL10	All the topics as listed below from Session 9 and Session 10 of <i>Server-side Development with NodeJS</i> book should be covered in <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 9 – Integration of Node.js with MongoDB</u> ➤ Describe the process to connect Node.js with MongoDB ➤ Demonstrate the CRUD operations on a MongoDB database using Node.js ➤ Explain the building of CRUD APIs in Node.js <u>Session 10 – Building Websites using Node.js</u> ➤ Explain the process of creating a Web application using Node.js ➤ Describe the steps to connect the Web application to a MongoDB cloud database ➤ List the steps to upload the Web application to the GitHub repository ➤ Explain the procedure to deploy the Web application on Render from a GitHub repository	<u>Server-side Development with NodeJS</u> SG – Session 9 & 10 XP – Session 9 & 10 TG – Session 9 & 10
11	NODEJS – TL11	All the topics as listed below from Session 11 and Session 12 of <i>Server-side Development with NodeJS</i> book should be covered in <i>Server-side Development with NodeJS</i> book should be covered in this session. <u>Session 11 – Node.js in Action (Application Development) - I</u>	<u>Server-side Development with NodeJS</u> SG – Session 11 & 12 XP – Session 11 & 12 TG – Session 11 & 12

Session No.	Session Title	Session Details	Deliverables' Mapping
		➤ Explain the process of creating a Web application using Node.js ➤ Describe the steps to connect the Web application to a MongoDB database <u>Session 12 – Node.js in Action (Application Development) - II</u> ➤ List the steps to upload the Web application to the GitHub repository ➤ Explain the procedure to deploy the Web application on Render from a GitHub repository	
12	NODEJS – TL12	The Try It Yourself questions from Session 6 to Session 12 of <i>Server-side Development with NodeJS</i> book should be covered in this session.	<u>Server-side Development with NodeJS</u> Session 6-12
	S4-S6	All the Assignments from <i>Session 6 to Session 12 of Server-side Development with NodeJS</i> course should be solved in this session.	<u>Server-side Development with NodeJS (OnlineVarsity)</u> Session 6-12

5. Library References

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| ➤ Node.js Web Development: Server-side web development made easy with Node 14 using practical examples, 5th Edition by David Herron |
| ➤ Node.js the Right Way: Practical, Server-Side JavaScript That Scales by Jim R. Wilson |

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